

Jordi Mursalim

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EDUCATION

University Of British Columbia

Bachelor of Applied Science, Computer Engineering

2024 – 2029

Vancouver, BC

EXPERIENCE

Software Engineering Intern - Vehicle Controls

May 2026 – December 2026

Rivian and Volkswagen Group Technologies

Vancouver, BC

- Validated safety-critical battery management system (state of charge/energy/health, contactor control, fault response) software by authoring 90+ SIL and HIL test cases supporting a program with 93% coverage across 97 software requirements.
- Designed a data-playback tool that records and replays SIL test execution to streamline the debugging process, exposing software component interactions for all test automation and firmware development domains.
- Blocked 8+ incorrect CAN signal definition changes from merging by creating an AI-integrated Git pipeline tool that diffs incoming signal definition edits against safety-critical signal specs and flags mismatches for review.
- Root-caused 16+ intermittent test failures to firmware and test-script defects by breaking down signal diagrams in Vector CANalyzer.

Mechatronics Team Lead

September 2025 – Present

UBC Baja SAE

Vancouver, BC

- Led an 8-member mechatronics sub-team inside a multi-disciplinary team of 70+ engineers, owning the data-acquisition and electrical systems for the 2026 Baja SAE competition vehicle
- Gathered actionable data to inform mechanical design iterations by delivering 5 custom electronic control units (ECU) and data acquisition units logging 18+ sensor channels.
- Engineered a wiring harness, ECU housing, and electronics package that sustained zero failures across 6+ hours of continuous off-road endurance testing by designing for vibration, shock, and dust resistance.

Software Developer Intern

September 2023 – June 2024

14 Oranges Software Inc.

Vancouver, BC

- Developed Bash scripts to reliably transfer client backups using SSH file sharing protocol. Automated backups up to 40,000 GB in size using the AWS CLI.
- Designed a scalable file transfer and storage system to streamline and secure backup management services for client-base of 150+ organizations, and 1000+ projects.

PROJECTS

Engine Data Acquisition | C, FreeRTOS, STM32 HAL, Hardware Diagnostic Tools

- Instrumented the powertrain with 8+ sensors, designing the end-to-end implementation of the wiring harness, electronics, and firmware. Analyzed sensor information to fine-tune engine performance and validate mechanical design predictions.
- Guaranteed deterministic sensor sampling on STM32 by architecting FreeRTOS C firmware using priority-assigned tasks and DMA-driven ADC reads. Verified concurrency and timing of sensor reading and telemetry tasks by analyzing real-time visualization of task execution and interrupts with SEGGER SystemView.
- Performed hardware and software debugging to validate behavior and data quality of sensors and peripheral units, eliminating 3 crucial sensor data-quality faults before competition.

Driver Dashboard | C++, QT, QML, Linux, Raspberry Pi

- Created an embedded C++ application for dynamically parsing a set of 25+ message definitions (CAN DBC file) to standardize decoding across every telemetry node.
- Built a QML front end rendering live speed, engine RPM, and sensor status signals, giving both the driver in-race feedback and the team information on the status of data acquisition systems.

TECHNICAL SKILLS

Hardware & Embedded: STM32 HAL, Arduino, Raspberry Pi, CAN, Hardware Diagnostic Tools, SIL/HIL

Developer Tools: Git, GDB, STM32 Toolchain, Vector CANalyzer, SEGGER SystemView, Linux CLI, Jira, Jenkins

Languages: C/C++ (FreeRTOS), Python, Java, Bash